

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 17



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Recap of Previous Lecture



Topic

Boolean algebra / Boolean lattice



Topic

Practice problems



Topics to be Covered



Topic

Functions



Topic

Domain / Range of a function



Topic

Injective (one-one) function



Topic

Surjective (onto) function



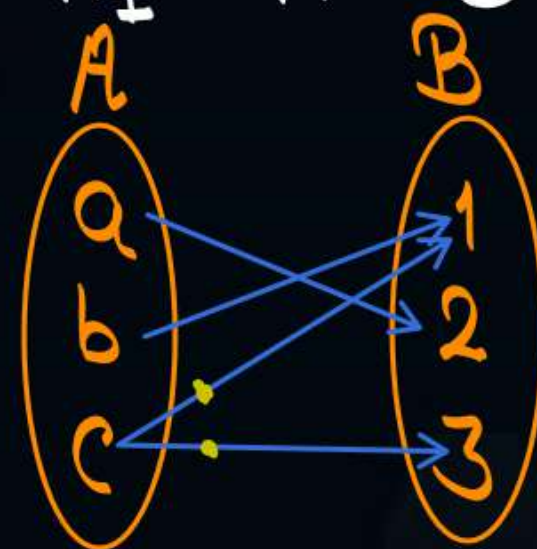


Topic : Function

→ Every function from Set A to Set B is also a relation from Set A to Set B, but Every relation from Set A to Set B need not be a function from Set A to Set B.

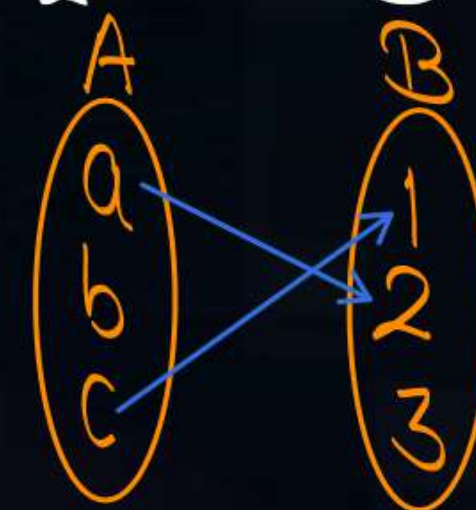
A relation from Set A to Set B is called a function from Set A to Set B if and only if Every element of Set A relates with exactly 1 element of Set B.

eg: $R_1: A \rightarrow B$ { R_1 is a relation }
from A to B

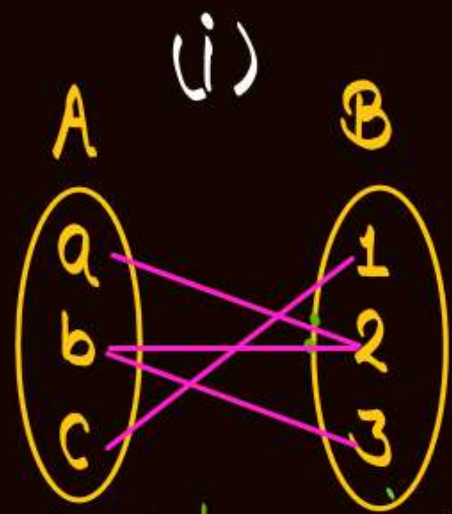


{ element 'c' of Set A relates with two elements of Set B.
∴ Not a function

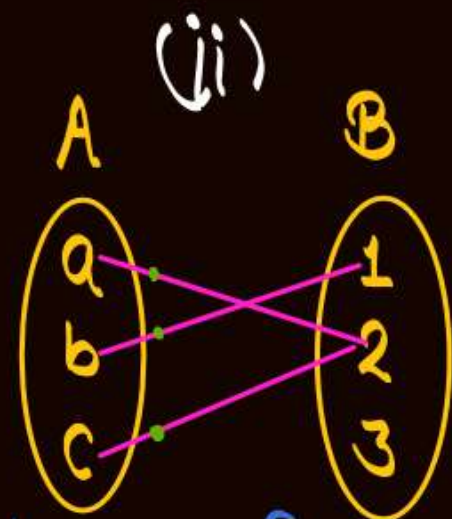
eg: $R_2: A \rightarrow B$ { R_2 is a relation }
from A to B



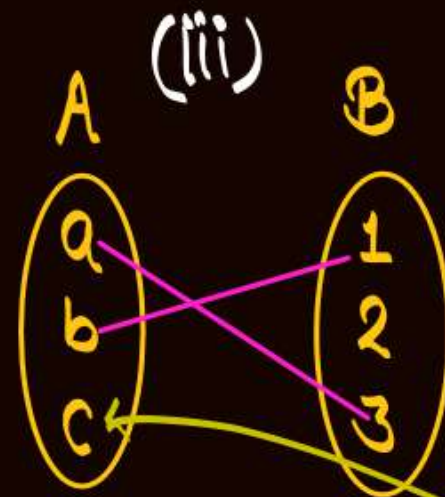
{ Element 'b' relates with '0' element of Set B.
∴ Not a function



'b' relates with two elements $\therefore$ Not a function from A to B

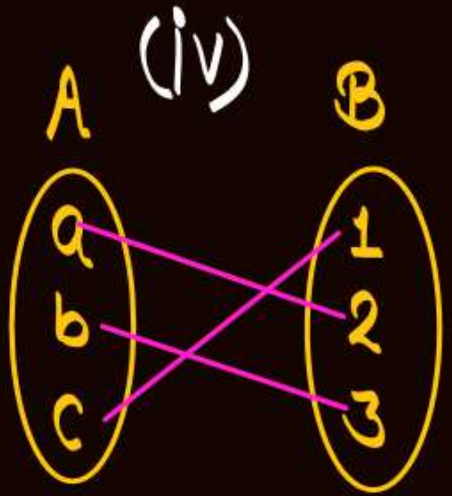


it is a function from set A to set B, but not a function from B to A

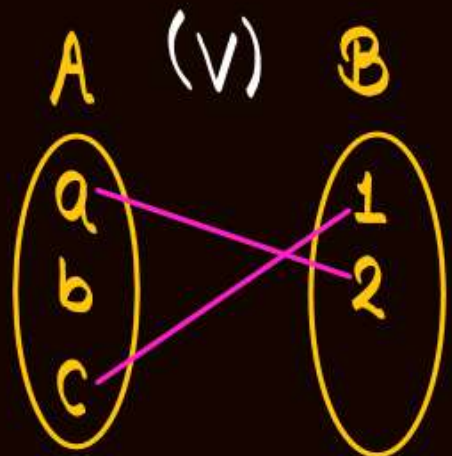


Neither a function from A to B, nor a function from B to A

'c' relates with '0' element



It is a function from A to B, as well as from B to A

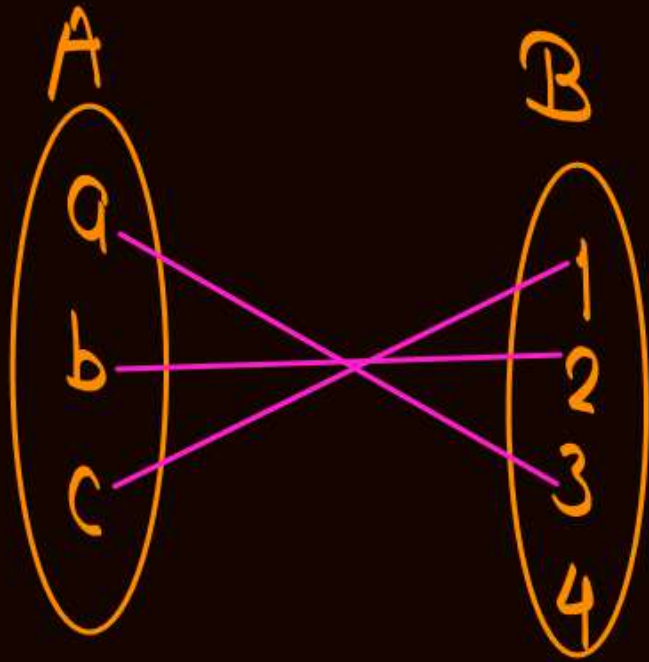


Not a function from set A to set B, but it is a function from B to A

→ Let $f: A \rightarrow B$ is a function,

then set 'A' is called domain of the function 'f'
and set 'B' is called Co-domain of the function 'f'

eg. $f: A \rightarrow B$



Here $f(b) = 2$.

then we say that

image of 'b' w.r.t. function 'f' is '2'
(or)

We say that 'b' is the pre-image
of element '2' w.r.t. function 'f'



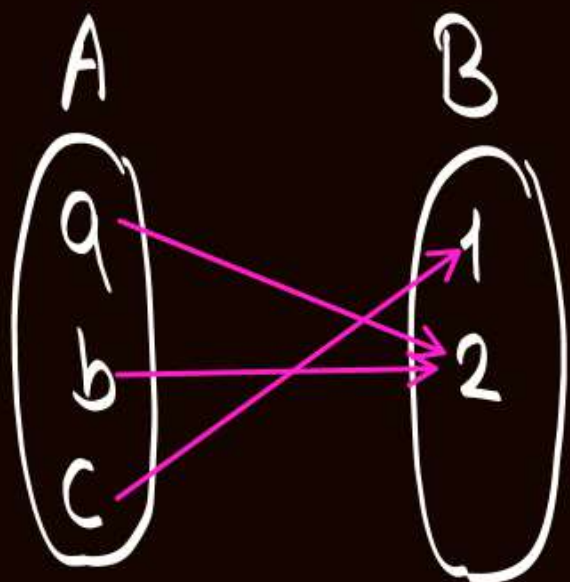
Topic : Range of a function

- ★ The set of all elements from Co-domain that are mapped by at least one element of domain is called "Range" of the function.

$$\text{Range} \subseteq \text{Co-domain}$$

eg:

$$f_1: A \rightarrow B$$



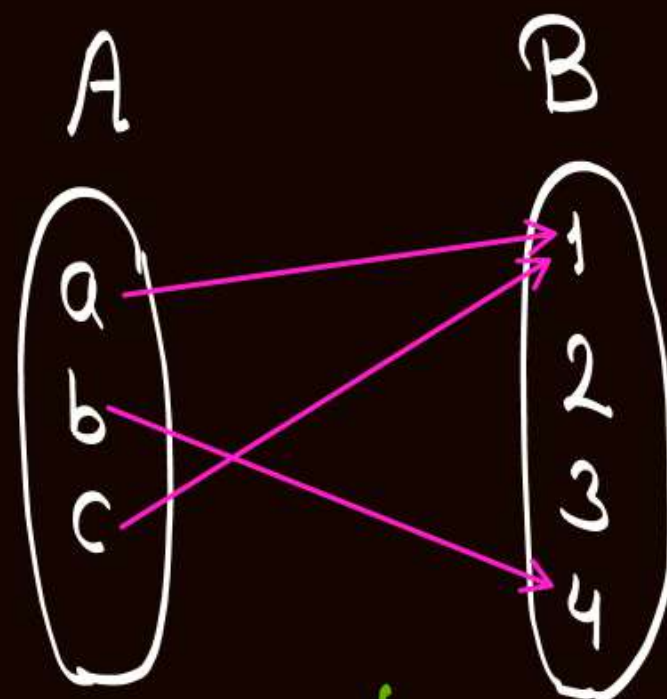
Domain: $\{a, b, c\}$

Co-domain: $\{1, 2\}$

Range = $\{1, 2\}$

Range = Co-domain

$$f_2: A \rightarrow B$$



Domain: $\{a, b, c\}$

Co-domain = $\{1, 2, 3, 4\}$

Range = $\{1, 4\}$

Range $\subset$ Co-domain



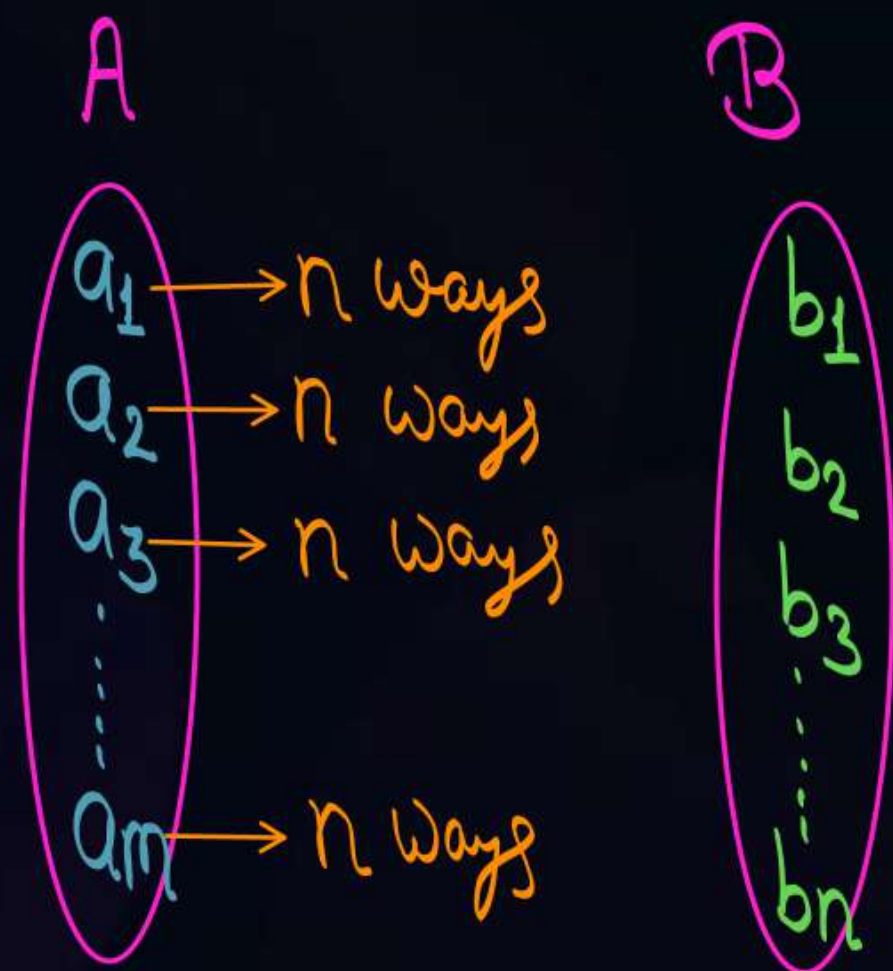
Topic : Total number of functions

* Let $|A|=m$ and $|B|=n$, then

Total Number of functions Possible from set A to set B = $|B|^{|A|} = n^m$

$$= (\text{size of Co-domain})^{(\text{Size of domain})}$$

m times n



$$\therefore \text{Total no. of functions Possible from A to B} = \underbrace{n * n * n * \dots * n}_{m \text{ times } n} = n^m$$

Note:-

A function defined from set A to set A itself is called a function on set A .

If $|A| = n$, then

Total no. of functions possible on set A _{from A to A } $= |A|^{|A|} = n^n$

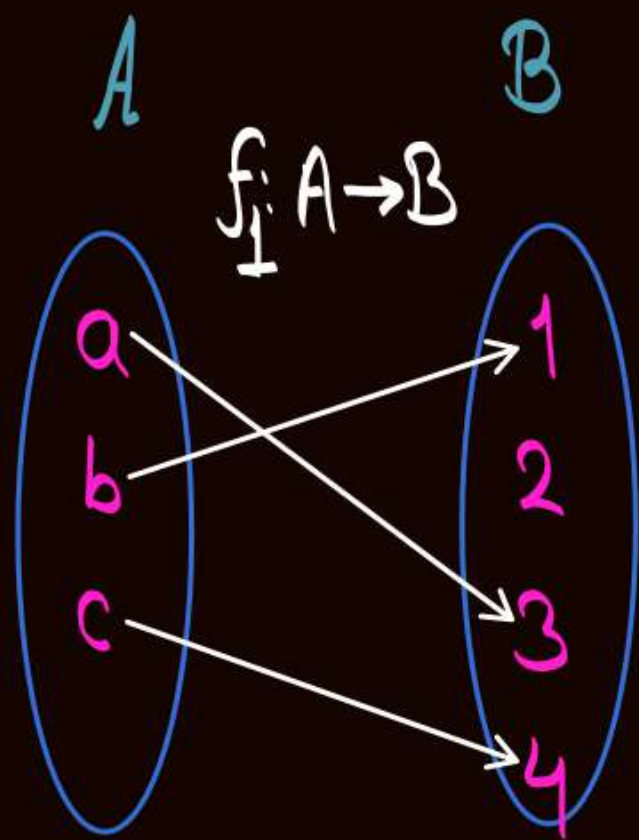


Topic : Injective (one-one) function

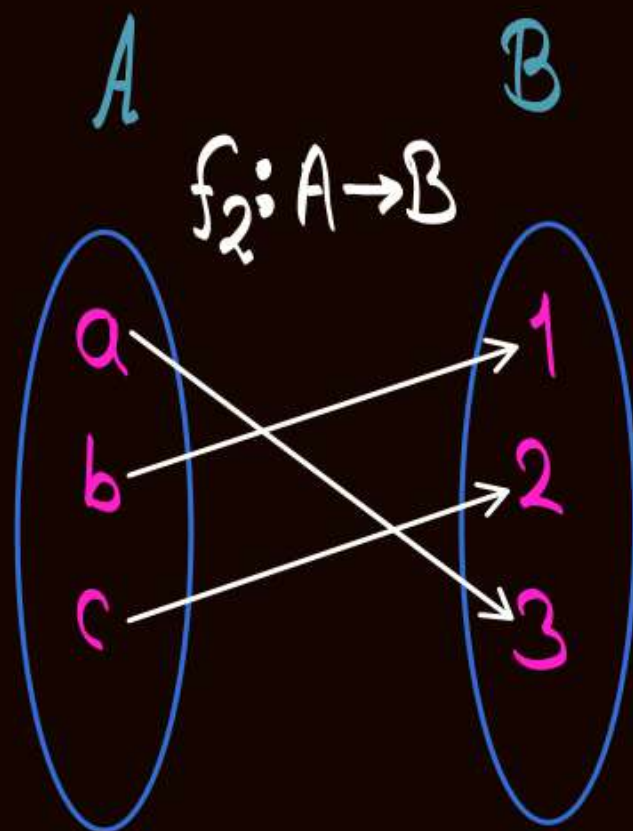
A function $f: A \rightarrow B$ is called an injective (one-one) function if and only if.

distinct elements of the domain (set A) have distinct images in Codomain (set B)

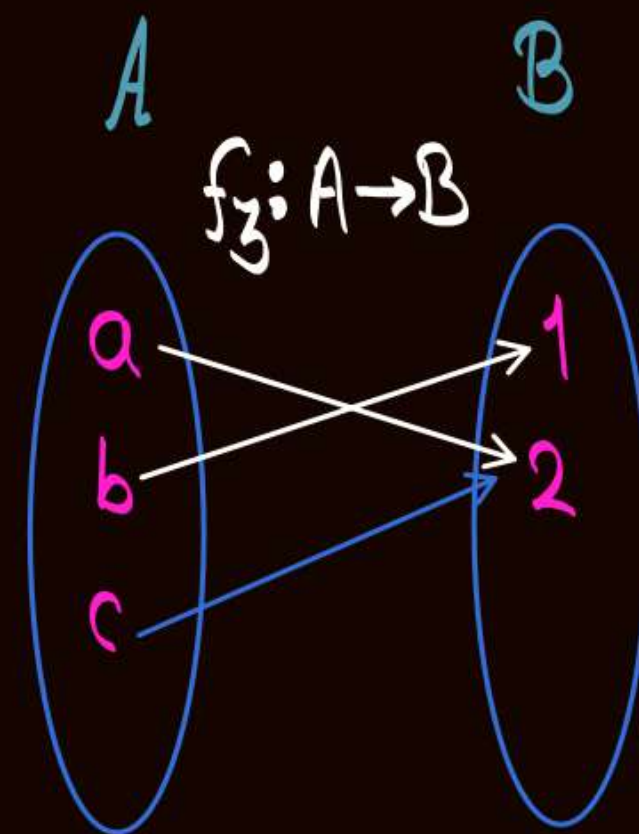
i.e. if $a, b \in \text{Domain}$
and $a \neq b$, then $f(a) \neq f(b)$ } $\forall a, b \in \text{Domain}$



" f_1 " is an
injective function
(one-one)



" f_2 " is an
injective function
(one-one)



$a \neq c$
 $f(a) = 2$ & $f(c) = 2$
 We have $f(a) = f(c)$
 even when $a \neq c$
 $\therefore$ " f_3 " is not an injective
 function

Note:-

① If $|A| > |B|$, then
One-one function is not possible from set A to set B

② If One-one function exists from set A
to set B, then $|A| \leq |B|$

③ In one-one function Every element of
Co-domain can have at most '1' pre-image



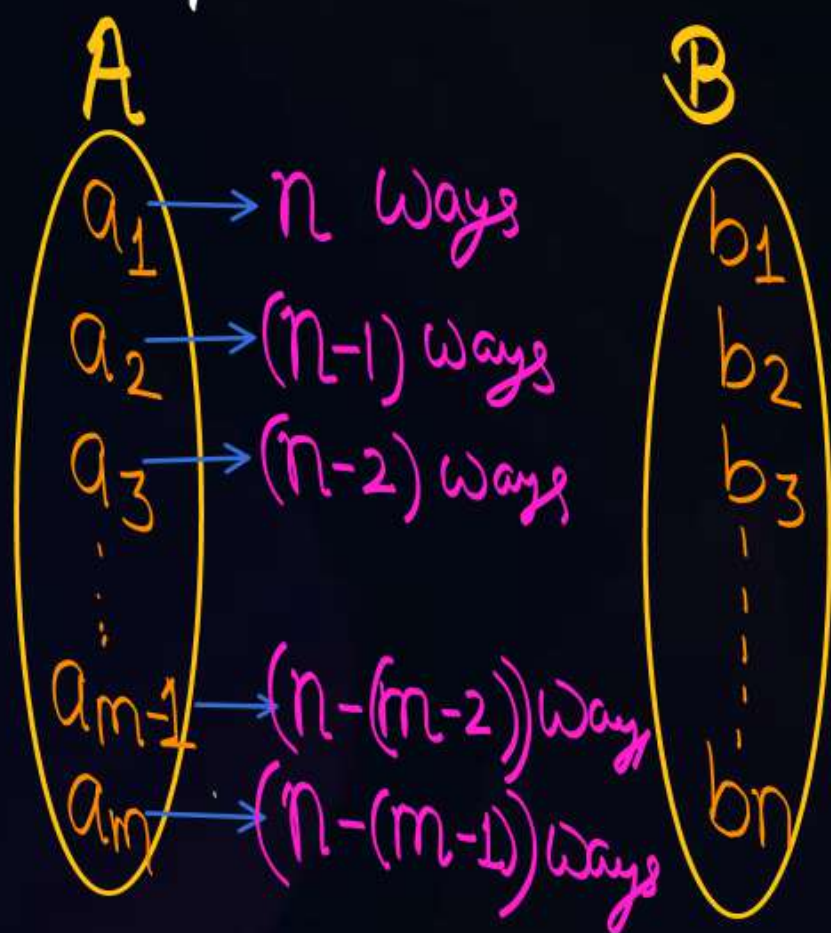
Topic : Number of one-one function

$${}^n P_r = \frac{n!}{(n-r)!}$$

Let $|A| = m$ & $|B| = n$ (such that $m \leq n$),

then,

Number of one-one functions possible from set A to set B = $\frac{n!}{(n-m)!} = {}^n P_m$



∴ Number of one-one functions possible from set A to set B

$$= n \times (n-1) \times (n-2) \times \dots \times (n-m+2) \times (n-m+1) \times \frac{(n-m)(n-m-1) \dots 3 \cdot 2 \cdot 1}{(n-m)(n-m-1) \dots 3 \cdot 2 \cdot 1}$$

$$= \frac{n!}{(n-m)!}$$

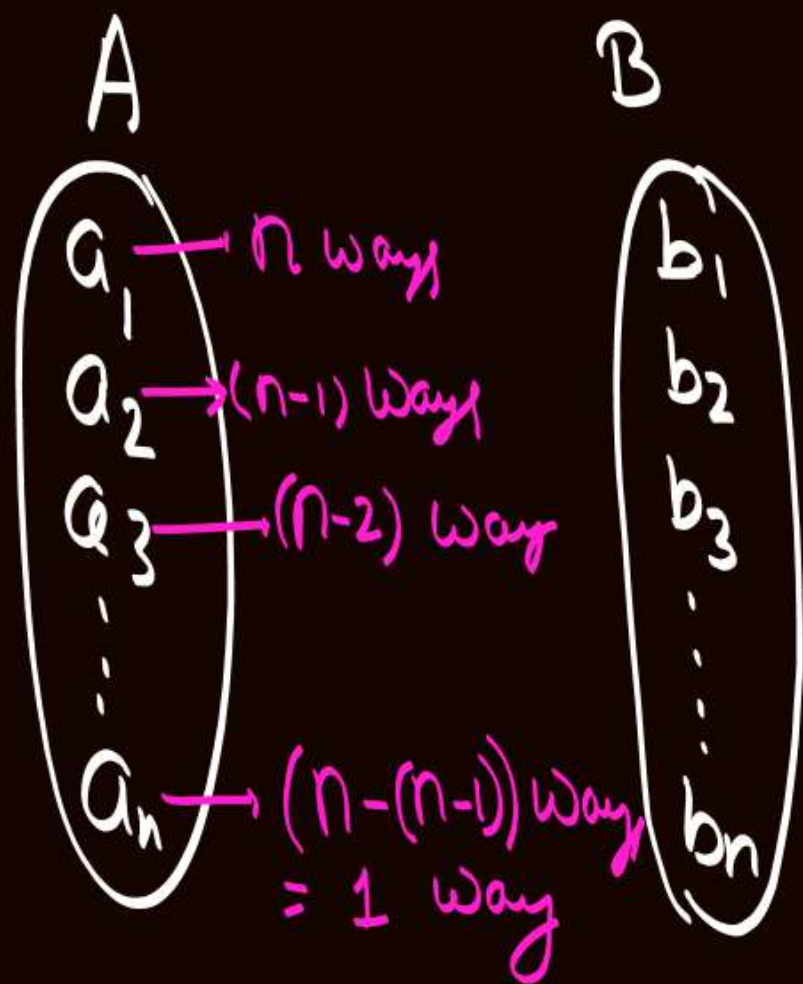
Note:

If $|A| = |B| = n$,

then, number of one-one function

Possible from set A to set B $= {}^n P_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$

$$0! = 1$$



$$\begin{aligned} \therefore \text{No. of one-one function possible} &= n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 3 \cdot 2 \cdot 1 \\ &= n! \end{aligned}$$



Topic : Surjective (onto) function

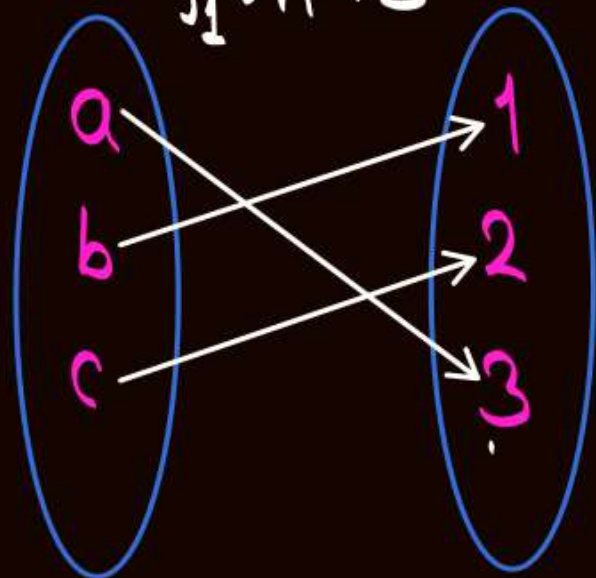
A function $f: A \rightarrow B$ is called an Onto (Surjective) function if and only if

Every element of Co-domain is mapped by at least one element of domain

i.e. A function is called onto function if and only if $\text{Range} = \text{Co-domain}$

$$|A| = |B|$$

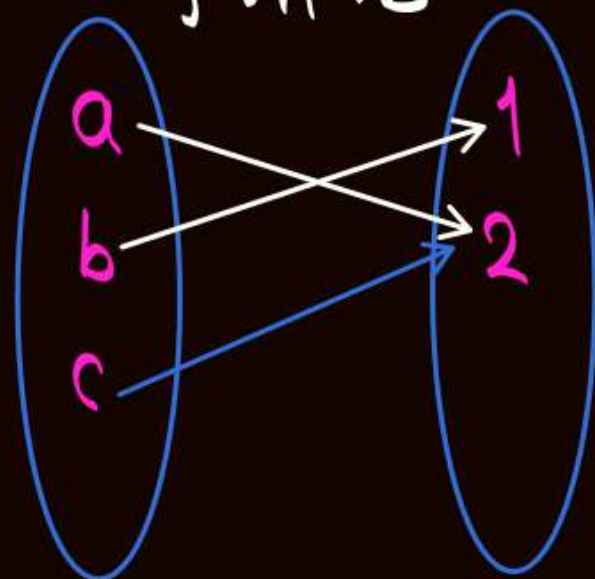
$$f_1: A \rightarrow B$$



Range = Co-domain
 $\therefore f_1$ is an
 Onto function

$$|A| > |B|$$

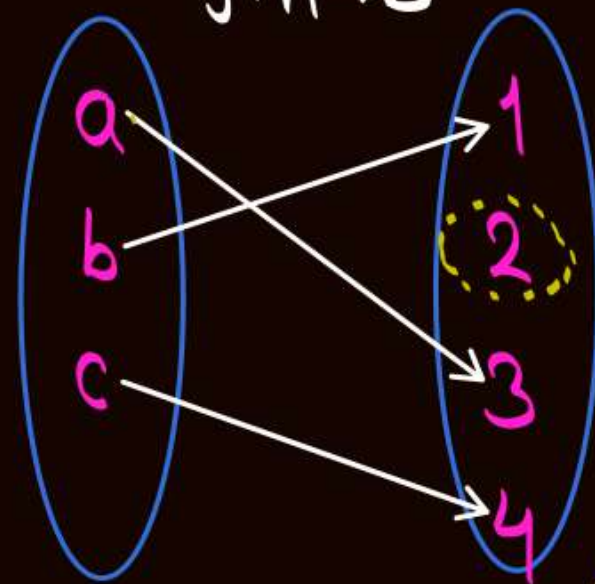
$$f: A \rightarrow B$$



Range = Co-domain
 $\therefore f_2$ is an
 Onto function

$$|A| < |B|$$

$$f: A \rightarrow B$$



Range $\subset$ Co-domain
 $\therefore f_3$ is not an
 Onto function

Range = $\{1, 3, 4\}$

Co-domain = $\{1, 2, 3, 4\}$

Note:

- ① If $|A| < |B|$, then
Onto function is not possible from set A to set B.
- ② If there exist an onto function from
set A to set B, then $|A| \geq |B|$
- ③ In an onto function Every element of
Co-domain will have at least '1' pre-image



Topic : Number of onto functions

① If $|A|=n$ & $|B|=2$, (such that $n \geq 2$) then
How many Onto functions are possible from A to B

② If $|A|=|B|=n$, then
How many Onto functions are possible from A to B

H.W.

③ if $|A|=4$ & $|B|=3$, then
How many onto functions are possible from A to B.

Q:- If $|A| = |B| = n$
then How many onto functions are possible from A to B.

Note: If $|A| = |B|$, then
Every one-one function from set A to set B will be Onto,
and Every onto function from set A to set B will be one-one.

i.e, When $|A| = |B|$, then

Number of one-one functions
from set A to set B = Number of onto functions
from set A to set B

* When $|A| = |B| = n$, then
No. of onto functions from A to B = No. of one-one functions from A to B = $n!$



2 mins Summary



Topic

Functions

Topic

Domain / Range of a function

Topic

Injective (one-one) function

Topic

Surjective (onto) function

THANK - YOU